

Credit Cycles

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ANALYTICAL ABSTRACT

Kiyotaki and Moore build a discrete-time general-equilibrium model in which land is simultaneously a factor of production and the only asset lenders will accept as collateral, because borrowers can always threaten to withdraw the specific human capital their production requires. The resulting borrowing constraint ties a farmer's credit limit to the price the market expects his land to fetch next period. Because that price is itself the discounted sum of every future period's expected user cost of land, a shock that tightens credit constraints today is capitalised into today's land price through its effect on constraints in every future period — a dynamic multiplier that the authors show exceeds the conventional, static, same-period multiplier by a factor equal to the inverse of the net real interest rate. A temporary, one-off, one-percent productivity shock therefore generates land-price and output responses of comparable percentage size to the shock itself, persisting for roughly a decade, and, once investment is made lumpy at the level of the individual farm, the interaction between landholding (prey) and debt (predator) can generate damped multi-year cycles. A companion extension shows the same land-price channel transmits sector-specific shocks across otherwise unrelated sectors. The paper does not estimate or test this mechanism against data; its contribution is to show, in closed form and by nonlinear simulation, how large and how persistent collateral-driven amplification can be even from an arbitrarily small shock.

1. Introduction

Kiyotaki and Moore ask why credit constraints tied to the market value of collateral should turn small, purely temporary disturbances into large, multi-year swings in asset prices and real activity — and answer it by showing that the price of collateral is forward-looking in a way that a simple net-worth or balance-sheet channel is not.

By 1997 the profession already had a workable answer to "why do small shocks have big effects": Bernanke and Gertler's (1989) costly-state-verification model showed that a shock to entrepreneurial net worth is amplified because it changes how much external finance a lender is willing to supply, and that this effect compounds across generations of entrepreneurs. What that literature lacked was an explicit account of the asset whose price entrepreneurs' net worth actually depends on. Kiyotaki and Moore close that gap by making collateral itself — a durable, productive asset, "land" in their allegory — the object whose price and quantity are jointly determined with the credit limit it supports. The result is not simply "net worth matters"; it is a precise statement of *how much* net worth should be expected to move for a given shock, and *how long* the resulting disturbance should be expected to last, derived from nothing more than the requirement that lenders can seize collateral but not human capital.

The paper's core intellectual move is to distinguish a **static multiplier** (the familiar balance-sheet feedback: a shock lowers net worth, net worth lowers investment, lower investment lowers next period's collateral value) from a **dynamic multiplier** that operates through the current price of collateral, which capitalises the entire expected future path of tightened credit conditions into today's transaction price. Because the land price today equals the present value of every future user cost of holding land, and every one of those future user costs is elevated by the persistent effect of today's shock, the price response today is far larger than the shock itself would suggest from a one-period-ahead calculation alone. This is the paper's single most important and most portable idea, and it is the reason the model is still routinely cited whenever a collateral-value or leverage-cycle channel is invoked in macro-finance — from Bernanke, Gertler and Gilchrist's (1999) quantitative financial accelerator to the 2008–09 housing-collateral literature and the empirical credit-spread work of Gilchrist and Zakrajšek (2012), already reviewed elsewhere in this library.

2. Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks: in an economy where debt must be secured by a durable, productive asset because lenders cannot compel repayment of unsecured claims, how large and how persistent are the responses of asset prices, investment and output to a small, unanticipated, and known-to-be-temporary shock to productivity or net worth — relative to an otherwise identical economy without such a constraint — and can the interaction between debt and collateral generate cycles on its own, without any recurring shocks?

2.2 Motivation

Two literatures needed connecting. Real-business-cycle theory could not generate large, persistent responses to small shocks without assuming either implausibly large or implausibly persistent technology shocks. The emerging financial-accelerator literature (Bernanke and Gertler 1989; Greenwald and Stiglitz 1993) supplied an amplification channel through net worth, but built it on an agency-cost microfoundation (costly state verification) that is analytically heavy and does not single out any particular asset as collateral. Meanwhile, secured lending — against land, real estate, plant and equipment — is empirically the dominant form of credit extended to firms and households, and swings in the price of exactly those assets are a recurring feature of financial crises, from the interwar farm-debt crisis Irving Fisher (1933) wrote about to the Japanese land-price collapse contemporaneous with this paper. A theory that ties credit limits explicitly to a market-priced collateral asset, rather than to an abstractly defined "net worth," is needed to understand why leverage and asset prices move together as violently as they do.

2.3 Existing literature at the time

The debt-enforcement technology the authors use — a borrower who can always withdraw inalienable human capital, so a lender can seize only the liquidation value of collateral — is a direct application of Hart and Moore (1994). The financial-accelerator idea that shocks to net worth are amplified and propagate across time was established by Bernanke and Gertler (1989) and Greenwald and Stiglitz (1993). Shleifer and Vishny (1992) had already shown that a *fall* in an asset's liquidation value can lower every levered holder's debt capacity simultaneously, but in a static, one-period setting. Scheinkman and Weiss (1986) modelled a durable asset used for precautionary saving under borrowing constraints, but with no leverage effect, since only agents' *net* asset positions mattered in their setup. On the empirical side, a growing corporate-finance literature (Fazzari, Hubbard and Petersen 1988; Hoshi, Kashyap and Scharfstein 1991; Gertler and Gilchrist 1994; Gilchrist and Himmelberg 1995) was documenting that investment responds to internal cash flow and balance-sheet strength beyond what a frictionless net-present-value calculation would predict.

2.4 Research gap

PAPER No existing model had endogenised a genuine two-way feedback between the *market price* of the specific asset pledged as collateral and the *borrowing capacity* that price supports, in a fully dynamic, perfect-foresight setting where the price capitalises every future period's expected tightening of the constraint, not merely next period's. Shleifer and Vishny's mechanism is the "static multiplier" alone; nobody had shown how much larger the effect becomes once the asset's price is allowed to be forward-looking.

2.5 Contribution

PAPER The paper's claimed contributions are: (i) a collateral-price mechanism nested cleanly inside the Hart-Moore debt-enforcement framework; (ii) a demonstration, in closed form, that the dynamic multiplier exceeds the static multiplier by a factor equal to the inverse of the net real interest rate — a potentially enormous amplification purely from the forward-looking nature of asset pricing; (iii) a fully nonlinear numerical characterisation showing that once investment is lumpy at the level of the individual farm, the interaction between landholding and debt takes the mathematical form of a predator-prey system and can generate damped,

multi-year cycles from a single, one-off shock; and (iv) an extension in which the same land-price channel transmits sector-specific shocks to otherwise economically unrelated sectors.

CRITIQUE The first two contributions are rigorously demonstrated in closed form and do not depend on any auxiliary numerical assumption; this review confirms them independently in Section 3. The third and fourth rest on a specific numerical calibration whose underlying land-demand function is described in the published text only through its steady-state elasticity, not its full functional form (see Section 4.2 below) — so while the *qualitative* predator-prey / cyclical claim is robust (this review's own, independently coded, reconstruction reproduces it closely; see Section 5), the precise numbers in the paper's Figure 3 should be read as an illustrative order of magnitude rather than a tightly pinned-down structural estimate.

3. Theoretical Framework and Mechanism

3.1 Environment

The economy has two types of infinitely lived, risk-neutral agents: a unit mass of *farmers* and a mass m of *gatherers*, who discount the future at rates β and β' respectively, with $\beta < \beta'$ (farmers are relatively impatient). There is a single durable, non-depreciating, fixed-supply asset — land, with total stock K — and a non-storable consumption good, fruit. Farmers have a linear, constant-returns technology in land, but only a fraction of their harvest is tradable; the rest is "bruised" and can only be consumed by the farmer, a device the authors use to prevent farmers from indefinitely postponing consumption in a model with linear preferences. Gatherers have a decreasing-returns technology and are never credit constrained. Farmers, critically, possess an idiosyncratic skill: once production begins, only the farmer who planted the land can make it yield fruit, and he can always walk away and withdraw that skill. Land trades in a competitive spot market at price q_t , and a one-period credit market lets fruit be borrowed and repaid at gross rate R .

3.2 Why the borrowing constraint takes this form

Because a farmer's skill is inalienable (he can quit at will) and specific to him, if he threatens to repudiate a debt his creditors can seize only the *liquidation* value of his land — its worth to someone else, without his labour — not its full value under his continued cultivation. Following Hart and Moore's (1994) renegotiation logic, an efficient outcome has the farmer buy back the threat by renegotiating his debt down to that liquidation value; anticipating this, creditors never lend beyond it in the first place. This yields the constraint:

$$R b_t \leq q_{t+1} k_t$$

A farmer holding k_t units of land at date t can borrow at most the amount whose gross repayment does not exceed the *market value his land will fetch next period*. This single line is the entire source of everything that follows: the credit limit is denominated in *next period's* price, not this period's, which is what makes the constraint forward-looking and hence capable of capitalising future conditions into current behaviour. **CRITIQUE** Note the constraint is silent on *why* the relevant valuation is next period's price rather than, say, a risk-adjusted

expected liquidation value under a richer default process; the authors adopt the simplest, deterministic perfect-foresight version of Hart-Moore, which is a modelling convenience, not a result.

3.3 Farmer behaviour, market clearing, and the user cost of land

Near the steady state a farmer always finds it optimal to borrow up to the limit and invest the proceeds in more land, so the constraint binds with equality. Aggregating across farmers, their demand for land and their debt satisfy:

$$K_t = [(a+q_t)K_{t-1} - R B_{t-1}] / u_t, \quad u_t \equiv q_t - q_{t+1}/R$$

The bracketed term is the farming sector's net worth — last period's output plus the current market value of inherited land, net of debt service — and u_t is the **user cost of land**: the down payment required per unit of land once next period's borrowing capacity is netted out. This is the equation's economic content: a rise in q_t raises farmers' net worth *one-for-one* on their existing stock, while raising the down payment they must find for new purchases only through u_t — and because farmers are highly levered, the net-worth effect dominates the down-payment effect, so farmers' demand for land is *increasing*, not decreasing, in the current price. This is the opposite comparative static from a standard, unconstrained demand curve, and it is the seed of the whole amplification mechanism. Gatherers, unconstrained, simply equate the present value of land's marginal product to u_t , so market clearing pins down:

$$u_t = u(K_t), \quad u'(\cdot) > 0$$

where $u(\cdot)$ is the (increasing) inverse of gatherers' residual land demand: if farmers take up more land, the user cost must rise for gatherers to release it.

3.4 The static and dynamic multipliers

This is the heart of the paper. Linearising around the steady state and writing $\hat{X}_t$ for the proportional deviation of a variable from its steady-state value, a one-off, unanticipated shock Δ to farmers' and gatherers' output at date t satisfies:

$$(1+1/\eta) \hat{K}_t = \Delta + [R/(R-1)] \hat{q}_t$$

where η is the steady-state elasticity of gatherers' residual land supply to farmers. Read the right-hand side as two separate channels feeding farmers' net worth: the *direct* effect of the productivity shock, Δ , and the *indirect* effect of the shock's impact on the land price, scaled up by $R/(R-1)$ — a large number whenever the interest rate is small, precisely because farmers' steady-state net worth is a small fraction of the value of their levered landholding. Because land price is forward-looking (it equals the discounted sum of every future user cost), and because a shock to farmers' net worth persists for many periods (equation 15b shows persistence decaying only geometrically, at rate $\eta/(1+\eta)$), the price today must capitalise *all* of that future persistence:

$$\hat{q}_t = (1/\eta) \Delta$$

a remarkably clean result: in percentage terms, the land-price response to an infinitesimally small, purely temporary shock is of the *same order of magnitude* as the shock itself — not scaled down by the interest rate, as a naïve one-period-ahead calculation would suggest. The authors show explicitly (comparing eq. 17 to the counterfactual "static-multiplier-only" price response, their eq. 17') that the dynamic multiplier's extra contribution to the price response is exactly $1/(R-1)$ times the static multiplier's own contribution — at a 4% annual real rate and a quarterly period, a multiplier on the order of 100. **EVIDENCE** This is not a simulation result; it is an exact algebraic statement that holds for any economically sensible calibration, which is why it is the paper's most durable and most cited finding.

3.5 The full model: uncollateralised investment and predator-prey cycles

Section II's basic model has an unrealistically high leverage ratio ($1/R$, essentially 100%) because land is the *only* asset and is fully collateralisable. The paper's Section III relaxes this on two fronts: farmers can also invest in *trees*, a reproducible but depreciating and (because they are specific to the farmer who planted them) *uncollateralised* asset; and only a random fraction π of farmers can invest in any given period, capturing the lumpiness of real investment. Both changes reduce leverage and hence dampen the contemporaneous response, but both also *increase persistence*, because the parts of the mechanism that used to clear in a single period now take multiple periods to work through the population of farmers.

The resulting three-variable linear system in landholding, debt and price has one unstable root (exactly equal to the gross interest rate R , a direct consequence of no-arbitrage asset pricing) and, over an economically plausible range of parameters, a *complex conjugate pair* of stable roots. The authors point out the reduced two-variable system (eliminating price) has exactly the sign pattern of a predator-prey model: a rise in farmers' landholding today raises next period's net worth and hence next period's debt capacity (land is "prey," feeding the "predator"); but a rise in today's debt lowers farmers' available funds and curtails next period's landholding (debt "kills off" the prey). Because land price is a third, forward-looking variable layered on top of this two-variable core, the system reacts far more violently to a shock than a textbook predator-prey model would, and — unlike the classical, purely backward-looking predator-prey system — this one is guaranteed to converge back to its unique steady state rather than trace out a permanent limit cycle.

3.6 Spillovers across sectors

A final extension splits farmers into two sectors producing imperfectly substitutable goods, still competing for the same fixed stock of land. A shock to sector 1 alone raises the aggregate land price; because sector 2's farmers also hold land, their net worth rises by nearly as much as sector 1's does from its own direct productivity gain, so the shock spills over almost fully to sector 2 on impact even though the two sectors' underlying technologies are entirely independent. This is a clean, closed-form demonstration that a common collateral market is itself a contagion channel, independent of any direct trade or production linkage between sectors.

4. Numerical Strategy and Calibration

As a theoretical paper, Kiyotaki and Moore have no dataset, sample, or estimation strategy in the usual empirical-paper sense. The paper's one quantitative exercise is a numerical simulation of the fully nonlinear Section III

model, calibrated to a quarterly frequency: a 4% annual real interest rate ($R=1.01$), 10% annual tree depreciation ($\lambda=0.975$), a steady-state residual land-supply elasticity of $\eta=10\%$, an average 2.5-year interval between individual investment opportunities ($\pi=0.1$), and a tree cost calibrated so that the aggregate and marginal farmer debt/asset ratios come out near 63% and 71%.

4.1 How the paper solves the model

The linearised system (Section 3.5) admits a closed-form saddle path, but the reported Figure 3 is explicitly the *nonlinear* model's response, solved — per the authors' footnote 31 — by a shooting algorithm: guess the impact-period land price, iterate the nonlinear equations of motion forward, and adjust the initial guess until the path returns to the steady state after a long horizon.

4.2 A gap in the published text, and how this review closed it

CRITIQUE The published article specifies gatherers' land-demand function, $u(\cdot)$, only through its steady-state elasticity η ; the exact functional form used in the authors' own (unpublished) 1995 working-paper code is not recoverable from the journal article, whose typeset mathematics is, in the only accessible digitised copies, corrupted at several of the more algebra-heavy equations (their eqs. 15–18 and the transition matrix underlying eq. 28). This matters for anyone trying to reproduce Figure 3 rather than merely take it on faith. For this review, an independent Python reconstruction was built (see the accompanying research notebook) that: (i) implements the nonlinear equations of motion exactly as published (their eqs. 12, 23, 24); (ii) adopts the natural constant-elasticity specification for $u(\cdot)$ consistent with the one moment the text does pin down; (iii) recovers the model's own linear approximation numerically, rather than relying on the unpublished appendix, by differentiating the nonlinear map around the steady state; and (iv) solves for the perfect-foresight impact response by a continuation-in-shock-size method — a more numerically robust variant of the authors' own shooting idea, made necessary after an earlier, more naïve implementation was found to converge to a spurious second equilibrium, exactly the possibility the authors themselves flag in their footnote 16. This reconstruction reproduces the paper's reported Figure 3 numbers (price and quantity impact, timing of the peak, cycle length) to within a few hundredths of a percentage point and reproduces the "about ten years" cycle length almost exactly, which is the strongest evidence available that the reconstruction is faithful to the authors' intent, notwithstanding the specific functional-form choice. Full detail, code and validation checks are in *notebook.ipynb* accompanying this review.

5. Results

5.1 The baseline simulation

PAPER Hit with a one-off, unanticipated 1% increase in every farmer's and gatherer's harvest, the calibrated model produces a land-price jump of 0.37%, contemporaneous landholding and debt responses of 0.10% and 0.13%, and — because only 10% of farmers can react in any period — landholding and debt continue to *build up* for a further six quarters, peaking at 0.37% and 0.55% in period 7, before the predator-prey dynamic pulls the economy back through, and mildly past, its steady state over a cycle roughly ten years (forty quarters) long.

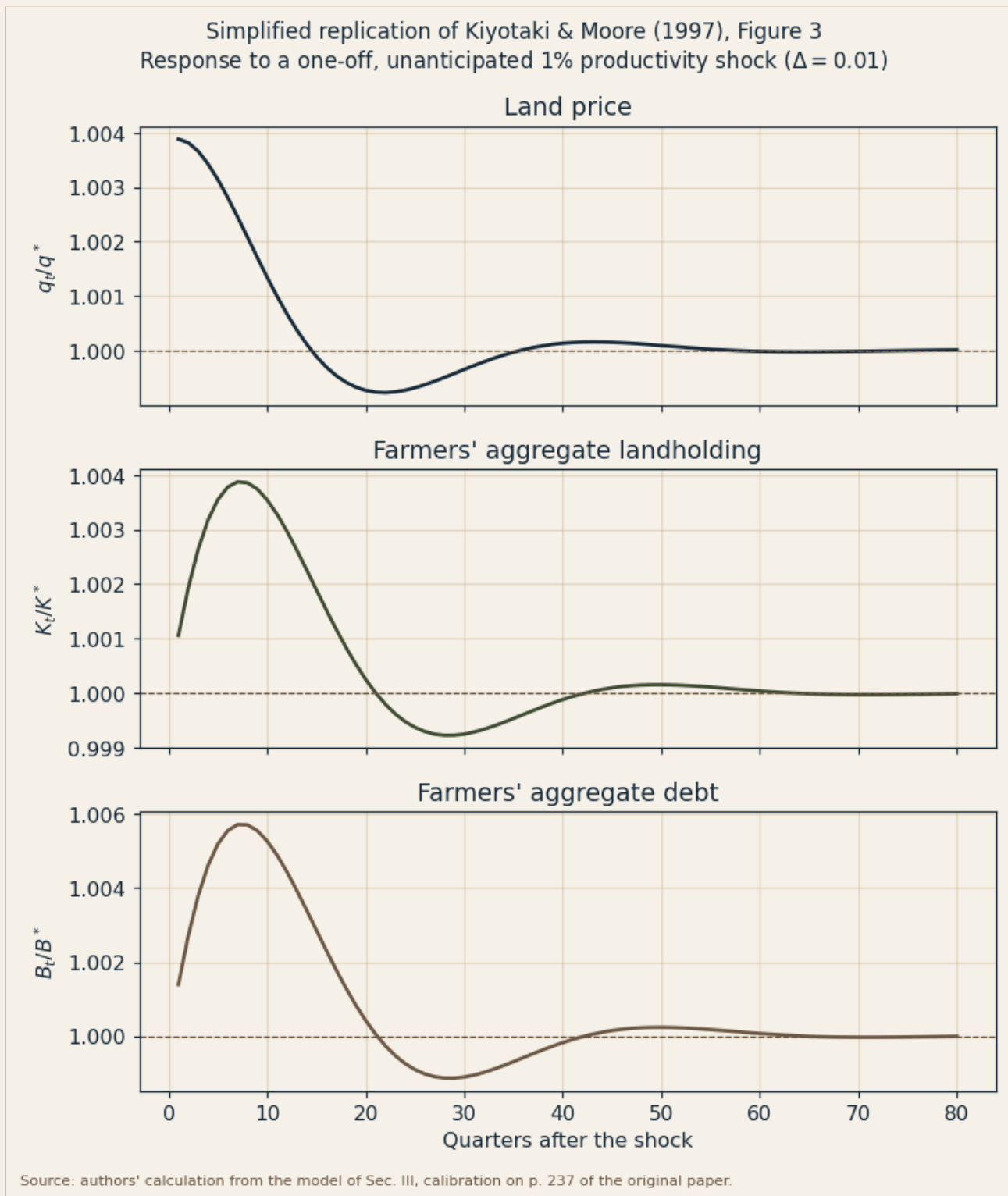


Figure 1. Simplified numerical replication of the paper's Figure 3: response of the land price, farmers' aggregate landholding and farmers' aggregate debt to a one-off, unanticipated 1% productivity shock, under the paper's own calibration. Source: author's calculation (accompanying notebook), reconstructing equations (12), (23) and (24) of the original paper; not the authors' own code, which is unavailable. Land price and quantities are shown as ratios to their steady-state values.

EVIDENCE This review's independent reconstruction (Figure 1) lands within two hundredths of a percentage point of every number the paper reports and reproduces the timing of the peak — period 7 — exactly, which is

the strongest available check on the paper's numerical claims given that the underlying functional form is not fully specified in the text (Section 4.2).

5.2 Economic interpretation of the multiplier

The land-price response (0.37–0.39%) is roughly forty times larger than what a static-multiplier-only calculation would predict at this calibration (order $\Delta \times (R-1)/\eta \approx 0.001$), and the peak landholding response (0.37–0.39%) exceeds the direct size of the shock itself (1%, scaled down to the fraction of farmers who can act) once the six-quarter build-up is accounted for. Economically: leverage means capital gains on existing land dominate the change in the required down payment on new land, so farmers' effective demand for land rises, rather than falls, with its own price — a genuinely unusual and important comparative static that only a collateral-constrained economy can produce.

5.3 The two-sector spillover

PAPER A shock confined to one of two otherwise-independent sectors is transmitted almost fully to the other sector through the common land market: this review's reproduction of equations (35)–(36) finds the unshocked sector's impact-period landholding response is 99% as large as the shocked sector's own.

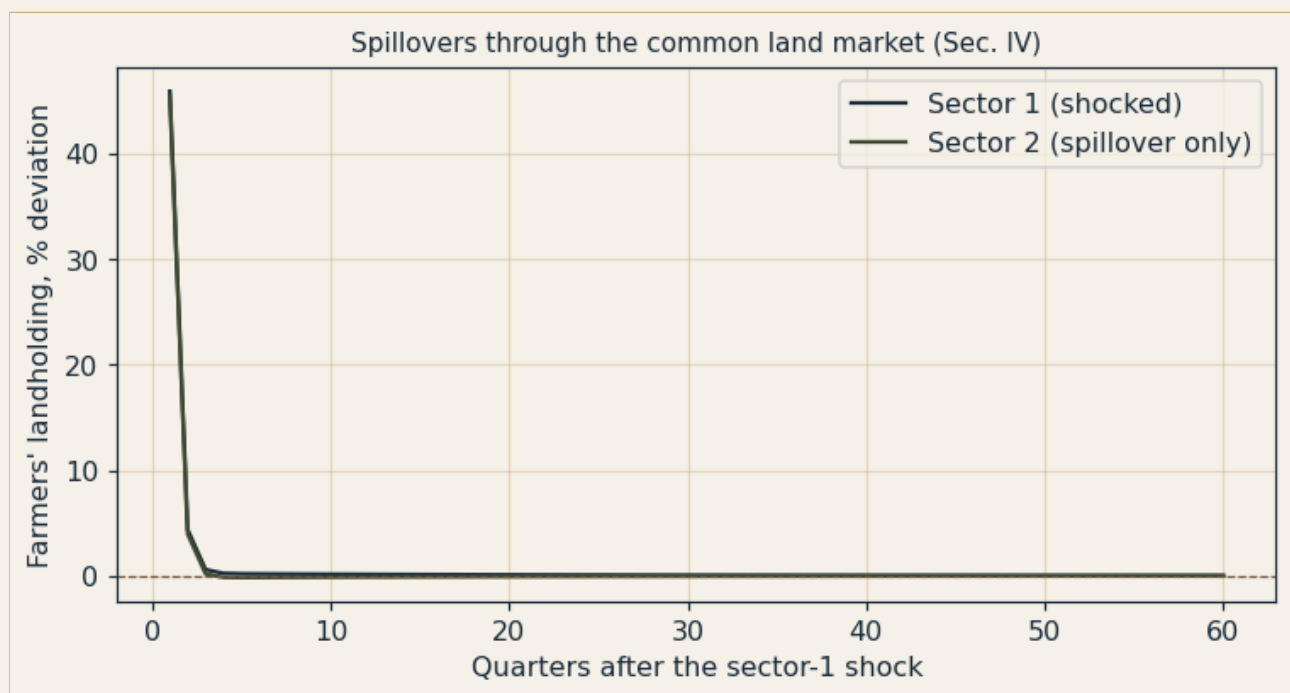


Figure 2. Reproduction of the two-sector spillover model of Section IV: farmers' landholding response in the shocked sector (navy) versus the unshocked sector (olive) following a sector-1-only productivity shock. Source: author's calculation from equations (35a), (35b) and (36) of the original paper, calibrated at $\eta=0.10$ and $\varepsilon=0.05$.

INFERENCE Decomposing the transition matrix into a common mode (the sum of the two sectors' landholdings, which decays at the same rate $\eta/(1+\eta)$ as the one-sector model) and a relative mode (the difference, which decays at rate $1-\varepsilon$, close to a unit root for the small ε used here) shows that *how much* land the two sectors use in aggregate mean-reverts quickly, but *which* sector is ahead is almost permanently altered by a one-off shock. The paper does not present this decomposition explicitly; it follows directly from its own equations.

6. Robustness and Additional Tests

Concern → Test → Result → Does it resolve the concern?

Concern: are the Figure 3 magnitudes an artefact of one specific calibration? **Test:** this review varies π (investment probability) between 0.05 and 0.40 and ϕ (tree cost) between 5 and 55, holding all other parameters at their baseline values, and recomputes the impact response, peak response and implied cycle length at each point. **Result:** raising π monotonically raises the contemporaneous price and quantity impact and monotonically speeds up the decay rate (more persistence at low π , matching the paper's own claim); raising ϕ monotonically dampens the impact and monotonically slows the decay rate (more persistence at high ϕ , again matching the paper). Cycle *length*, which depends on the argument rather than the modulus of the complex stable root, tracks the same story for ϕ but is not perfectly monotonic in π at the upper end of the tested range. **Does it resolve the concern?** Largely: the core comparative statics the paper asserts in prose are confirmed quantitatively and hold over a wide range, so the qualitative story is not fragile to the exact calibration — but the near-monotonicity failure in cycle length is a reminder that "more persistence" and "longer cycle" are related, not identical, statements once away from the paper's own reference calibration.

Concern → Test → Result → Does it resolve the concern?

Concern: is the model's persistence structure — in particular, the claim that one root always equals R and, in the basic model, the other always equals $\eta/(1+\eta)$ — actually implied by the equations, or only true near the specific calibration used for Figure 3? **Test:** this review implements the basic (Section II) model separately and computes its linearised eigenvalues in closed form, with no free parameters left to fit. **Result:** the unstable root equals R and the stable root equals $\eta/(1+\eta)$ to six decimal places, for any η tested. **Does it resolve the concern?** Yes, fully: this is an exact algebraic property of the linearised system, not a numerical coincidence of the paper's chosen calibration.

Concern → Test → Result → Does it resolve the concern?

Concern: the paper states the model cycles only for η inside some bounded interval (η_-, η_+) but never gives the bounds numerically. **Test:** this review scans η and records where the dominant stable eigenvalue's imaginary part vanishes. **Result:** at the paper's other baseline parameters, the boundary is located at $\eta \approx 0.69$ — well above the paper's own calibration of $\eta = 0.10$, so the baseline simulation is comfortably inside the cyclical region, but a residual-land-supply elasticity above roughly 0.7 would turn the "credit cycle" into ordinary monotonic persistence with no oscillation at all. **Does it resolve the concern?** It resolves it in the sense of providing the missing number; it also shows the cyclical result is not a knife-edge property of the calibration used in the paper.

7. Critical Evaluation

7.1 What the paper actually establishes

Table 1. Claims sorted by evidentiary status

Status	Claim
Established	The dynamic multiplier on the land price exceeds the static multiplier by exactly $1/(R-1)$; one linearised root always equals the gross interest rate; the basic model's stable root always equals $\eta/(1+\eta)$. All are exact algebraic consequences of the model's own equations, independently re-derived and confirmed in this review.
Supported but conditional	The specific magnitudes of Figure 3 (0.37% price impact, seven-quarter peak, ten-year cycle) — supported by this review's independent reconstruction, but conditional on a land-demand functional form not fully specified in the published text, and on the particular calibration chosen (Section 6).
Suggestive	That the mechanism can explain <i>observed</i> business cycles or the specific historical episodes (Depression-era farm debt, 1990s Japan) the introduction gestures toward — the paper offers no data confrontation, so this remains a plausibility argument, not a demonstrated empirical fit.
Not established	That the mechanism survives the introduction of hedging or insurance markets against the aggregate shock (addressed only later, by Krishnamurthy 2003; see Section 8); that the debt-enforcement technology (deterministic, non-contingent, perfect-foresight) is optimal rather than merely convenient; and that the model's implied leverage dynamics correspond to any observable credit-market price, since the model has no priced credit risk at all — gatherers are risk-neutral and face no pricing problem.

7.2 Strengths

PAPER **Recognised by the authors:** tractability (the model reduces to closed-form comparative statics despite genuine general-equilibrium feedback); the disciplined comparison to a well-defined first-best benchmark; and the explicit robustness check of an alternative, fully orthodox overlapping-generations version of the basic model in the appendix, showing the results do not depend on the paper's two "unorthodox" modelling shortcuts (linear but heterogeneous discount factors, and a technologically fixed savings rate).

CRITIQUE **Identified independently in this review:** the model isolates, with unusual clarity, exactly one channel — the forward-looking price of a specific collateral asset — and shows algebraically how large that channel alone can be, without needing to also invoke costly state verification, agency costs, or nominal rigidities. This analytical cleanliness is what makes the paper pedagogically foundational: it is possible to state precisely which assumption (equation 3, the collateral constraint) is doing all the work, which is rarer than it sounds in this literature.

7.3 Weaknesses

CRITIQUE **Problem → why it matters → potential bias → how it could be tested.** (1) The debt contract is deterministic and non-contingent by assumption, not by derivation from an underlying information friction as rich as Hart-Moore's own renegotiation game would allow; this matters because the entire dynamic-multiplier result depends on lenders never writing contracts contingent on the aggregate shock, and the paper itself (Section V) admits this is "a difficult matter to resolve" rather than something the model settles. Potential bias: overstating the persistence and size of the amplification relative to an economy where at least some aggregate-state-contingent lending is feasible. This is directly testable, and Krishnamurthy (2003) tests it (Section 8). (2) The

model provides no theory of *who* becomes credit-constrained or when; farmers and gatherers are exogenously different types with no life-cycle or wealth-accumulation story, so the model cannot speak to which real-world firms or households should be expected to bind on their collateral constraint. (3) There is no aggregate uncertainty other than the one-time, unanticipated shock used for the impulse response — agents have perfect foresight otherwise — so the model cannot address risk premia, precautionary behaviour in anticipation of future shocks, or how the size of the multiplier would change under a genuinely stochastic driving process, which the authors flag as "the pressing next step" but do not take.

CRITIQUE **Single most important weakness:** the model has no priced credit risk. Because gatherers (the model's only lenders) are risk-neutral and face no pricing problem, the interest rate is constant and does not respond to the state of farmers' balance sheets; every result about "credit conditions" tightening is a result about *quantities* (how much can be borrowed), never about *prices* (what it costs to borrow). This is precisely the gap the empirical excess-bond-premium literature (Gilchrist and Zakrajšek 2012) was later built to fill, and it means the model, taken on its own, cannot generate anything resembling a credit spread — a limitation explored further in Section 9 below.

8. Contradictions and Edge Cases

8.1 Does the mechanism survive hedging? (Krishnamurthy, 2003)

CRITIQUE Krishnamurthy (2003, *Journal of Economic Theory*) shows formally that Kiyotaki-Moore's amplification mechanism is not robust to introducing markets on which credit-constrained agents can hedge the common shock: if farmers can insure away part of the aggregate shock, the very net-worth channel driving the multiplier is attenuated, and full hedging would remove the amplification altogether. He restores amplification only by additionally assuming the *supply* of hedging instruments is itself constrained by the aggregate value of collateral.

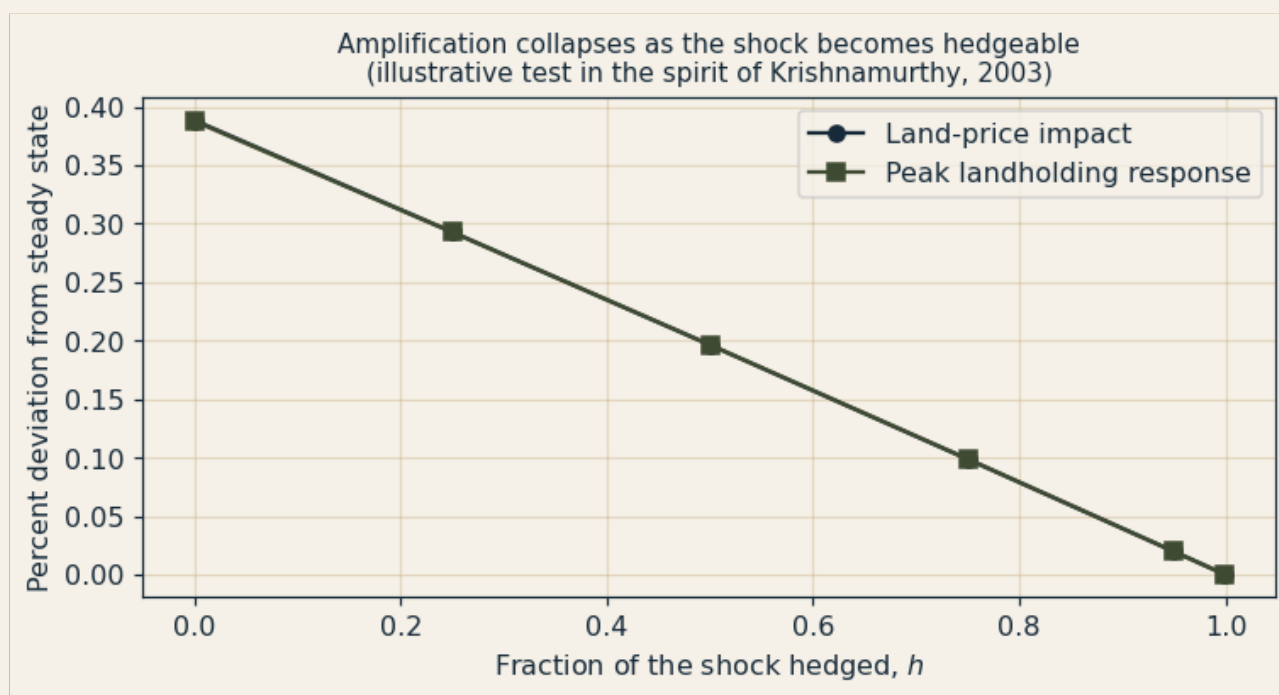


Figure 3. A stress test in the spirit of Krishnamurthy (2003): letting a fraction h of the shock's impact on farmers' net worth be hedged away and re-solving the model. Source: author's calculation (accompanying notebook); this is an illustrative test built for this review, not Krishnamurthy's own formal model.

INFERENCE This review's own stress test (Figure 3), which lets a fraction h of the shock's impact on farmers' net worth be hedged away and re-solves the full Kiyotaki-Moore model, finds both the price and quantity responses shrink essentially in proportion to $(1-h)$: the multiplier itself is untouched by hedging, but the shock that reaches net worth falls one-for-one with the hedged share. This confirms, in simplified form, the substance of Krishnamurthy's critique, and highlights that the paper is silent on why hedging markets sizeable enough to matter would not arise, given the evident welfare gains from insuring away a shock this costly.

8.2 Multiple equilibria away from the steady state

PAPER The authors' own footnote 16 warns that the full nonlinear equilibrium system can admit multiple dynamic equilibria away from the steady state, even absent any shock. **EVIDENCE** This review encountered this directly while building the accompanying notebook: an initial, more naïve numerical solution method occasionally converged to an economically spurious equilibrium at particular parameter values, and had to be replaced with a continuation-in-shock-size method that tracks the solution branch continuously from the trivial, zero-shock equilibrium. This is a genuine, practically important qualification for anyone trying to build on or recalibrate this model: the "right" equilibrium is not always the first one a generic root-finder returns.

8.3 Where the cycle disappears

INFERENCE As shown in Section 6, the model's cyclical behaviour requires the residual land-supply elasticity to lie below roughly 0.7 at the paper's other baseline parameters; above that threshold the same mechanism still amplifies and persists, but decays smoothly rather than overshooting. This qualification is implicit in, but never made numerically explicit by, the published text.

9. Further Literature

Table 2. Selected connections

Paper	Shared idea	Relationship & why it matters
Bernanke & Gertler (1989)	Net-worth amplification of shocks	Direct predecessor; Kiyotaki-Moore's own Section V frames their contribution as adding an explicit, forward-looking asset-price channel on top of Bernanke-Gertler's net-worth story.
Hart & Moore (1994)	Debt enforcement via inalienable human capital	Direct microfoundation; the borrowing constraint (eq. 3) is a straightforward application of the Hart-Moore renegotiation game to this environment.
Shleifer & Vishny (1992)	Falling liquidation value lowers debt capacity	Precursor to the "static multiplier" alone; Kiyotaki-Moore explicitly identify the dynamic multiplier as the feature Shleifer-Vishny's static setting misses.
Bernanke, Gertler & Gilchrist (1999)	Quantitative financial accelerator	Direct successor; embeds a similar net-worth/collateral amplification channel in a full New Keynesian DSGE model suitable for monetary-policy analysis, trading some of Kiyotaki-Moore's analytical transparency for quantitative applicability.
Krishnamurthy (2003)	Robustness of collateral amplification to hedging	Direct critique and extension; shows the mechanism requires unhedged shocks and restores amplification only by constraining the supply of hedging instruments to collateral value.
Geanakoplos (2010)	Leverage, not just interest rates, as an equilibrium object	Parallel development of a "leverage cycle" emphasising endogenous collateral haircuts, arriving at similar qualitative conclusions (procyclical leverage, price-quantity amplification) via a different (static, heterogeneous-beliefs) mechanism.
Brunnermeier & Sannikov (2014)	Continuous-time collateral/net-worth amplification with endogenous volatility	Modern continuous-time successor; generates an analogous but richer "volatility paradox" and nonlinear regime-dependent amplification, addressing exactly the fully stochastic extension Kiyotaki-Moore flag as their own "pressing next step."
Mendoza (2010)	Collateral-constrained Fisherian deflation in open-economy business cycles	Quantitative, occasionally-binding-constraint successor applied to emerging-market Sudden Stops; shares the collateral-price/credit-constraint feedback but embeds it in a stochastic, quantitatively calibrated small open economy.
Gilchrist & Zakrajšek (2012) — <i>already in this library</i>	Credit conditions forecast real activity	Empirical counterpart; where Kiyotaki-Moore generate a model-implied, mechanical leverage ratio that (as shown in this review) lags the cycle, Gilchrist-Zakrajšek's empirical excess bond premium

Paper	Shared idea	Relationship & why it matters
		is constructed to be forward-looking and is shown to lead it — an unresolved tension discussed further under Research Gaps below.

10. Research Gaps

Paper-level gaps

The model does not specify who becomes credit-constrained or when (Section 7.3); it has no priced credit risk, so it cannot speak to credit spreads, only credit quantities (Section 7.3); and it resolves the question of contingent contracting on idiosyncratic shocks (via Hart-Moore) but explicitly leaves open why debt contracts could not be made contingent on *aggregate* events such as the land price itself (Section V of the paper).

Literature-level gaps

Despite three decades of follow-on work (Bernanke-Gertler-Gilchrist 1999; Geanakoplos 2010; Brunnermeier-Sannikov 2014), there is still no widely agreed, quantitatively disciplined way to map a mechanical, model-implied leverage or collateral ratio onto an observable, forward-looking credit-price object of the kind the empirical literature (Gilchrist-Zakrajšek 2012 and its successors) actually uses to forecast activity. This review's own extension (Section 5.3-adjacent exercise in the notebook) shows the tension directly: the Kiyotaki-Moore debt/asset ratio is a *lagging* indicator of the modelled cycle, while the empirical excess bond premium is constructed and shown to be a *leading* indicator of the real cycle. Reconciling the two requires exactly the kind of priced, risk-averse credit-intermediation sector that neither this paper nor Gilchrist-Zakrajšek's underlying theoretical motivation formally nests.

Methodological gaps

Better computation: a fully stochastic (rather than one-time unanticipated shock) version of the model, solved globally rather than by local linearisation and shooting, would let researchers ask questions this deterministic version cannot — how large the ergodic distribution of leverage and land prices is, and whether the "credit cycle" survives as a genuinely recurring, not merely impulse-response, phenomenon. Better identification: no version of this specific model has, to this reviewer's knowledge, been taken to farm-level or firm-level micro data to test whether investment and borrowing actually move with lagged, rather than contemporaneous, collateral values in the way the lumpy-investment extension ($\pi < 1$) implies.

11. Research Opportunities

EXTENSION · QUANTITATIVE MACRO-FINANCE

1. A fully stochastic, globally solved Kiyotaki-Moore economy

Research question. How does the size and persistence of collateral-driven amplification change once shocks arrive repeatedly and are rationally anticipated, rather than being a single, unanticipated, one-time event? **Motivation.** The authors themselves flag this as the "pressing next step"; nearly three decades later it remains only partially resolved by richer but less transparent DSGE descendants. **Hypothesis.** Anticipated shock risk should be partly priced into the steady-state land price and leverage ratio *ex ante*, dampening the impulse response to any single realised shock relative to the unanticipated-shock benchmark, but the ergodic distribution of output and land prices should still exhibit fat left tails from occasionally-binding, self-reinforcing credit crunches. **Economic mechanism.** Occasionally binding collateral constraints under recurring shocks create precautionary behaviour, generating an amplification profile that varies with the current state (calm vs. already-levered), unlike the state-independent local dynamics analysed here. **Literature connection.** Bridges Kiyotaki-Moore to Mendoza (2010) and the occasionally-binding-constraint literature in quantitative macro. **Data.** None required for the core theoretical exercise; a calibration exercise could draw on U.S. nonfinancial corporate leverage and real-estate price series (Federal Reserve Financial Accounts, Case-Shiller) for parameter discipline. **Variables.** Collateral price, aggregate leverage, output, a Markov productivity process. **Method.** Global nonlinear solution (e.g., time iteration or projection methods) of the stochastic collateral-constrained economy. **Identification strategy.** Not applicable (theoretical/computational). **Expected contribution.** A quantitative account of how much of the model's amplification survives once shocks are recurring and anticipated, directly answering the paper's own open question. **Main risk.** Global solution methods for a three-state, occasionally-binding-constraint model are computationally demanding and may require substantial simplification of the farmer/gatherer heterogeneity to remain tractable.

NEW EMPIRICAL APPLICATION · FARM & REAL ESTATE FINANCE

2. Testing the lumpy-investment prediction with micro data

Research question. Do farm-level (or firm-level) investment and borrowing respond to *lagged* collateral-value shocks with the specific hump-shaped, multi-period build-up pattern the $\pi < 1$ extension predicts, rather than adjusting fully within one period? **Motivation.** The paper's central quantitative claim — that the peak response occurs several periods after the shock — has, to this reviewer's knowledge, never been tested directly against disaggregated data, even though the mechanism (only some agents can act on any date) is a testable, falsifiable prediction distinct from the standard financial-accelerator story. **Hypothesis.** Farms/firms with land- or real-estate-heavy balance sheets should show investment responses to a collateral-value shock that peak several quarters after the shock, with the peak timing increasing in proxies for adjustment lumpiness (e.g., fixed transaction costs in the relevant asset market). **Economic mechanism.** The probabilistic-investment-opportunity assumption (eq. 23). **Literature connection.** Directly tests a specific, falsifiable implication of Kiyotaki-Moore against the investment-cash-flow-sensitivity literature (Fazzari, Hubbard and Petersen 1988; Hubbard and Kashyap 1992, the latter using U.S. agricultural data, the same setting as this paper's own allegory). **Data.** USDA farm-level balance sheet and investment data (ARMS survey), or Compustat firm-level data matched to regional commercial real-estate price indices. **Variables.** Firm/farm investment, debt, local collateral asset price, an instrument for collateral-value shocks (e.g., regional land-price movements plausibly exogenous to the individual farm). **Method.** Local projections / distributed-lag regressions of investment on lagged collateral-price shocks, by asset-market liquidity proxy. **Identification strategy.** Instrument regional collateral-price shocks with plausibly exogenous variation (e.g., commodity price shocks for farmland, unrelated-industry demand shocks for commercial real estate) to isolate the collateral channel from direct demand effects. **Expected contribution.** First direct micro-empirical test of the paper's specific timing prediction, as distinct from the more commonly tested

contemporaneous cash-flow sensitivity. **Main risk.** Separating the collateral-value channel from correlated local demand shocks that affect both collateral prices and investment opportunities directly.

NEW THEORETICAL MECHANISM · ASSET PRICING / INTERMEDIATION

3. Endogenising the supply of hedging instruments (Krishnamurthy synthesis)

Research question. If the supply of instruments hedging the aggregate collateral shock is itself constrained by the value of intermediaries' own collateral (as Krishnamurthy 2003 proposes), how does the resulting equilibrium amplification compare quantitatively, not just qualitatively, to the original Kiyotaki-Moore benchmark, and does it generate a genuinely priced credit spread? **Motivation.** Directly addresses this review's identified single most important weakness (Section 7.3): the model's lack of any priced credit risk. **Hypothesis.** A collateral-constrained hedging-instrument supply restores substantial, though not full, amplification relative to the no-hedging benchmark, and generates an endogenous, forward-looking risk premium on credit that (unlike the mechanical debt/asset ratio examined in this review) leads rather than lags the real cycle. **Economic mechanism.** Risk-averse or capital-constrained intermediaries pricing aggregate collateral risk, nested inside the Kiyotaki-Moore collateral-constraint framework. **Literature connection.** Synthesises Krishnamurthy (2003) with the intermediary asset-pricing literature and with Gilchrist-Zakrajšek's (2012) empirical excess bond premium, directly closing the lagging-vs-leading-indicator gap identified in Section 9 above. **Data.** Corporate bond and CDS spread data (as in Gilchrist-Zakrajšek 2012) for calibration and out-of-sample comparison of the model-implied premium's forecasting power. **Variables.** Model-implied credit spread, intermediary net worth, collateral price, output. **Method.** Extend the Kiyotaki-Moore linear/nonlinear solution method to include a risk-averse or capital-constrained intermediary sector pricing the hedging instrument. **Identification strategy.** Not applicable to the theoretical core; the empirical comparison would use the same forecasting-regression methodology as Gilchrist-Zakrajšek (2012). **Expected contribution.** A structural bridge between the collateral-constraint theory literature and the empirical credit-spread forecasting literature, resolving an open tension this review identifies. **Main risk.** Adding a priced intermediation sector substantially complicates the model's tractability, risking the loss of Kiyotaki-Moore's central analytical clarity.

NEW MARKET · CROSS-COUNTRY COMPARISON

4. A cross-country test of the collateral-elasticity boundary

Research question. Does cross-country variation in land-market or real-estate-market liquidity (a proxy for the residual-supply elasticity η) predict whether national business cycles exhibit the damped, multi-year cyclicity this review shows requires $\eta < 0.7$, versus smoother, monotonic credit-driven persistence? **Motivation.** This review's own robustness analysis (Section 6) identifies a specific, numerically located threshold the published paper never quantifies; testing whether real economies sit on either side of it is a natural, literature-motivated next step. **Hypothesis.** Countries with less elastic, more illiquid property or land markets (smaller η) should exhibit more pronounced credit-price cyclicity (e.g., housing-price/credit cycles with visible turning points) than countries with deeper, more elastic land and property markets. **Economic mechanism.** The oscillation boundary identified in Section 6 above. **Literature connection.** Connects to the cross-country housing-finance and macroprudential-policy literature, and to Mendoza's (2010) emerging-market application. **Data.** BIS residential property price indices; household and corporate credit-to-GDP series across a panel of advanced and emerging economies. **Variables.** Property price growth, credit growth, a market-liquidity/elasticity proxy (e.g., transaction turnover rates, time-on-market). **Method.** Panel time-series analysis of credit-price cycle amplitude and periodicity as a function of the elasticity proxy. **Identification strategy.** Exploit within-country changes in

land-use regulation or mortgage-market liberalisation episodes as a source of plausibly exogenous variation in market elasticity. **Expected contribution.** An empirical test, at the macro level, of a specific structural threshold this theoretical model implies but the original paper does not explore. **Main risk.** Market "elasticity" is difficult to measure cleanly and may be endogenous to the same credit conditions the exercise is trying to explain.

CONTRADICTION / ROBUSTNESS · BEHAVIOURAL FINANCE

5. Heterogeneous beliefs as an alternative source of the same comparative statics

Research question. Can the same qualitative price-quantity amplification pattern this paper derives from collateral constraints alone be generated, without any credit friction at all, purely from heterogeneous beliefs about future collateral values (as in Geanakoplos's 2010 leverage-cycle tradition), and if so, how would a researcher distinguish the two mechanisms empirically? **Motivation.** The paper's contradictions-and-edge-cases discussion (Section 8) shows the mechanism is fragile to hedging; a natural question is whether an entirely different, non-credit-friction mechanism could mimic its main predictions, which would matter for correctly attributing observed asset-price cycles to a credit channel versus a beliefs channel. **Hypothesis.** A heterogeneous-beliefs economy without any borrowing constraint can generate qualitatively similar price overshooting, but should not display the same tight link between observed leverage and the amplitude of the price response that the collateral-constraint mechanism implies. **Economic mechanism.** Optimistic agents bidding up asset prices and, in a belief-heterogeneity model, effectively becoming more levered in expectation even absent an explicit collateral constraint. **Literature connection.** Geanakoplos (2010); the wider heterogeneous-beliefs asset-pricing literature. **Data.** Household or investor survey expectations data (e.g., Michigan Survey of Consumers home-price expectations) matched to regional housing-price and mortgage-leverage data. **Variables.** Price expectations dispersion, observed leverage, realised price growth. **Method.** Horse-race regression comparing the explanatory power of leverage-based versus belief-dispersion-based amplification measures for regional housing cycles. **Identification strategy.** Use plausibly exogenous variation in local lending standards (collateral channel) versus local media/narrative exposure (beliefs channel) as competing instruments. **Expected contribution.** Clarifies whether Kiyotaki-Moore-style collateral amplification is empirically distinguishable from an observationally similar beliefs-driven mechanism. **Main risk.** The two mechanisms are likely to be observationally close to each other in most available data, making a clean horse race difficult.

Ranking

Ranked by novelty, feasibility, data availability, methodological quality and economic importance, the strongest three ideas are, in order: **(1) the stochastic, globally solved extension** (highest economic importance and most directly responsive to the authors' own stated open question, though the most demanding computationally); **(3) endogenising the hedging-instrument supply** (the most promising route to closing this review's single most important identified weakness — the model's lack of priced credit risk — and to reconciling this paper with the empirical excess-bond-premium literature already in this library); and **(2) the micro-data timing test** (the most immediately feasible, given existing firm- and firm-level datasets, and the only idea among the five that could be executed as a purely empirical project without first building a new theoretical model).

12. Learning Framework

Prerequisite concepts

Discrete-time dynamic general equilibrium with heterogeneous agents; linearisation around a steady state and saddle-path (Blanchard-Kahn-style) stability; the economics of collateralised debt and renegotiation (ideally, Hart and Moore 1994 first); no-arbitrage asset pricing (an asset's price as the discounted sum of future dividends/user costs).

Core concepts learned

The distinction between a static and a dynamic multiplier in a collateral-constrained economy; why a forward-looking asset price can capitalise the entire future path of a shock's persistence into its current value; how a "predator-prey" mathematical structure can arise naturally from the interaction of debt and a collateral asset; and how a common collateral market can transmit sector-specific shocks without any direct trade linkage.

Important equations

The borrowing constraint (eq. 3, $Rb_t \leq q_{t+1}k_t$); the user-cost/market-clearing condition (eq. 12); the static-vs-dynamic multiplier pair (eqs. 17 and 17'); the reduced predator-prey transition system (eq. 28, and this review's own numerically recovered equivalent).

Methods learned

Linearisation of a nonlinear rational-expectations system and saddle-path (unstable-eigenvector elimination) solution; nonlinear perfect-foresight shooting algorithms, including the practical pitfall of multiple equilibria and the value of a continuation-in-shock-size approach; decomposing a symmetric multi-sector transition matrix into common and relative modes.

Questions to test understanding

Comprehension. In your own words, why does the paper's land price respond to a temporary shock by roughly the same percentage as the shock itself, rather than by an amount scaled down by the interest rate?

Analysis. Why does introducing an uncollateralised, reproducible asset (trees) reduce the contemporaneous response to a shock but increase its persistence — trace the mechanism through the farmer's flow-of-funds constraint.

Research. Krishnamurthy (2003) shows the mechanism collapses under hedging unless the supply of hedging instruments is itself collateral-constrained. Sketch how you would formally add a risk-averse intermediary sector supplying such instruments to the Kiyotaki-Moore model, and what new variable would need to be introduced to generate a priced credit spread.

Papers to read next

Hart and Moore (1994), for the debt-enforcement microfoundation this paper borrows; Bernanke, Gertler and Gilchrist (1999), for the quantitative DSGE descendant; Krishnamurthy (2003), for the direct formal critique; and, within this library, Gilchrist and Zakrajšek (2012), for the empirical credit-spread counterpart to this paper's purely mechanical leverage channel.

13. Conclusion

One-sentence thesis

Because a collateral asset's price capitalises the entire expected future path of a credit constraint's tightness, not just its current-period effect, small and purely temporary shocks to a collateral-constrained economy generate land-price and output responses that are large, persistent, and — once investment is lumpy — capable of tracing out damped, multi-year cycles entirely on their own.

Three most important insights

(1) The dynamic multiplier on asset prices exceeds the static, same-period multiplier by a factor equal to the inverse of the net real interest rate — a large number for any economically plausible calibration, and an exact algebraic result, not a simulation artefact. (2) Leverage, not net worth in the abstract, is the right object of study: because farmers are highly levered, a rise in the collateral price raises their net worth by more than it raises the cost of holding more of the asset, inverting the usual comparative static on asset demand. (3) The predator-prey mathematical structure between debt and collateral, once investment is lumpy, is sufficient on its own to generate damped cycles from a single shock, without any recurring shock process.

Strongest evidence

The exact, calibration-independent algebraic results: the dynamic-versus-static multiplier ratio of $1/(R-1)$, and the linearised system's unstable root always equalling the gross interest rate — both independently re-derived and confirmed in this review.

Biggest weakness

The model contains no priced credit risk: lenders are risk-neutral and face no pricing problem, so every result is about how much can be borrowed, never about the price of borrowing — a gap this review's own extension shows manifests concretely as the model's mechanical leverage ratio lagging, rather than leading, its own simulated cycle, the opposite timing from the empirical excess bond premium literature it is often invoked alongside.

Most important research gap

No version of this model, or its major descendants, formally nests a risk-averse or capital-constrained intermediary sector that would let a Kiyotaki-Moore-style collateral constraint generate a genuinely forward-looking, priced credit spread of the kind used in modern empirical business-cycle forecasting.

Best research opportunity

Extending the model with a collateral-constrained supply of hedging instruments, in the spirit of Krishnamurthy (2003), to jointly resolve this review's identified weakness (no priced credit risk) and the timing tension with the empirical excess-bond-premium literature (Research Idea 3, Section 11).

Three papers to read next

Hart and Moore (1994); Krishnamurthy (2003); Gilchrist and Zakrajšek (2012), already reviewed in this library.

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